Traffic-light controller + mod-N counter

Complete synchronous digital design document | Moore FSM and mod-6 counter

Design assumptions

The traffic system has two crossing roads, A and B. One transition occurs per externally selected timing interval (green duration or yellow duration). There are no sensor inputs; therefore the behavior is a fixed, safe cyclic sequence. Outputs are active-high lamp enables: G = green, Y = yellow, R = red.

1. Two-road traffic light - Moore finite-state machine

A Moore implementation is selected because each lamp pattern depends only on the registered state. The controller therefore changes lights only immediately after a clock edge, rather than as a direct response to a changing input.

State definition and assignment

State	Meaning	Q1 Q0	Road A	Road B
S0	A has right of way	00	Green	Red
S1	A clearance interval	01	Yellow	Red
S2	B has right of way	10	Red	Green
S3	B clearance interval	11	Red	Yellow

Cycle: S0 -> S1 -> S2 -> S3 -> S0. Reset forces S0, which is a safe starting display (A green, B red).

State/next-state/output table

Present	Q1Q0	Next	D1D0	AG AY AR	BG BY BR
S0	00	S1	01	1 0 0	0 0 1
S1	01	S2	10	0 1 0	0 0 1
S2	10	S3	11	0 0 1	1 0 0
S3	11	S0	00	0 0 1	0 1 0

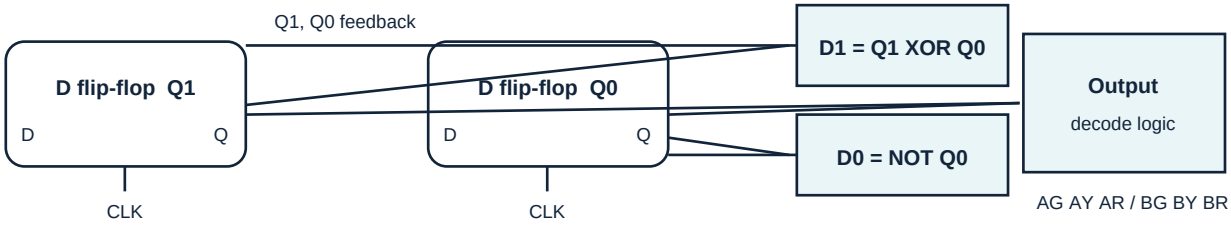
For D flip-flops, each D input equals the desired next-state bit.

K-maps and minimized equations

	Q0=1
	1
	0

D0	Q0=0	Q0=1
Q1=0	1	0
Q1=1	1	0

Next state: D1 = Q1 XOR Q0; D0 = NOT Q0. **Outputs:** AG = NOT Q1 · NOT Q0; AY = NOT Q1 · Q0; AR = Q1; BG = Q1 · NOT Q0; BY = Q1 · Q0; BR = NOT Q1.



Both flip-flops share the timing-interval clock. Reset initializes S0 = 00 (Road A green).

2. Synchronous mod-6 counter

Choice: N = 6. Three D flip-flops encode the six legal binary states. The counter moves 000, 001, 010, 011, 100, 101, then returns to 000 on the next shared clock edge.

Complete next-state table

Present Q2Q1Q0	Next D2D1D0	Role
000	001	count 0 -> 1
001	010	count 1 -> 2
010	011	count 2 -> 3
011	100	count 3 -> 4
100	101	count 4 -> 5
101	000	terminal wrap
110	000	unused-state recovery
111	000	unused-state recovery

K-maps / implementation equations

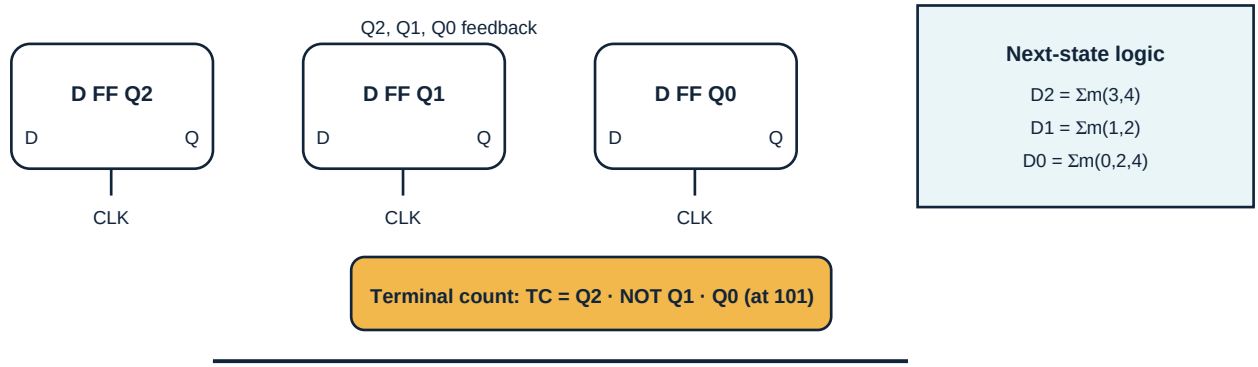
The following minterm equations come directly from the table (variables ordered Q2 Q1 Q0). They are valid for all eight present states, including self-recovery from the two unused states.

D2 = Σm(3,4) = NOT Q2 · Q1 · Q0 + Q2 · NOT Q1 · NOT Q0
D1 = Σm(1,2) = NOT Q2 · (NOT Q1 · Q0 + Q1 · NOT Q0)
D0 = Σm(0,2,4) = NOT Q0 · (NOT Q2 + NOT Q1)

Q2Q1Q0	00	01	11	10
0	D2/D1/D0: 0/0/1	0/1/0	1/0/0	0/1/1
1	1/0/1	0/0/0	0/0/0	0/0/0

Combined 3-variable K-map values are listed as D2/D1/D0; Gray-code columns are 00, 01, 11, 10.

All three D flip-flops are clocked together; logic makes 101 advance synchronously to 000.



Terminal-count logic and wrap-around proof

Terminal count: TC = Q2 · NOT Q1 · Q0. This expression is 1 only for 101 (decimal 5), the final legal count. Substitution into the next-state equations at Q2Q1Q0 = 101 gives D2 = 0, D1 = 0, and D0 = 0. Therefore, on the next rising clock edge, Q2Q1Q0 becomes 000. Since every prior legal row advances to its binary successor, the sequence visits exactly six states before repeating: 000 -> 001 -> 010 -> 011 -> 100 -> 101 -> 000. This proves synchronous mod-6 wrap-around.

Reflection: Moore versus Mealy

I chose a Moore FSM because the traffic lamps are safety-critical visible outputs: tying them only to registered state makes each complete light pattern stable for the entire timing interval. A Mealy machine could respond one clock period sooner to an input, but it would also allow input changes or logic hazards to affect the lamps between clock edges. With a fixed timing controller, the Moore design is simpler to verify, produces clean transitions, and maps naturally to an all-red/clearance-safe state sequence.